4.2 Adding, Subtracting, and Multiplying Polynomials

Name _____ Period ____ Date _____

Algebra II 4.2 (30-minute plan) · Student sheet

I can…

- add and subtract polynomials.
- multiply polynomials, including three binomials.
- use special product patterns and Pascal's Triangle.

Vocabulary · fill in the missing word

Like terms — terms with the same _____ raised to the same powers

Square of a binomial — $(a \pm b)^2$ = _____

Sum and difference — $(a + b)(a - b)$ = _____

Cube of a binomial — $(a \pm b)^3$ = _____

Pascal's Triangle — each number is the sum of the two above it; row n gives the _____ of $(a + b)^n$

Part 1 · Notes: fill in the blanks

To subtract, distribute the minus sign to _____ term.

$(a + b)^3$ = _____

Row n of Pascal's Triangle gives the coefficients of _____.

Part 2 · Examples (work them with the class)

Example 1

Find the product: $(x - 2)(3x^2 + x - 4)$

Example 2

Find the product: $(x + 1)(x - 3)(x + 2)$

Part 3 · Partner practice

Partner practice

Find the product: $(x + 3)(x - 1)(x - 2)$